

Auditory AR System to Induce Pseudo-Haptic Force Feedback for Lateral Hand Movements Using Spatially Localized Sound Stimuli

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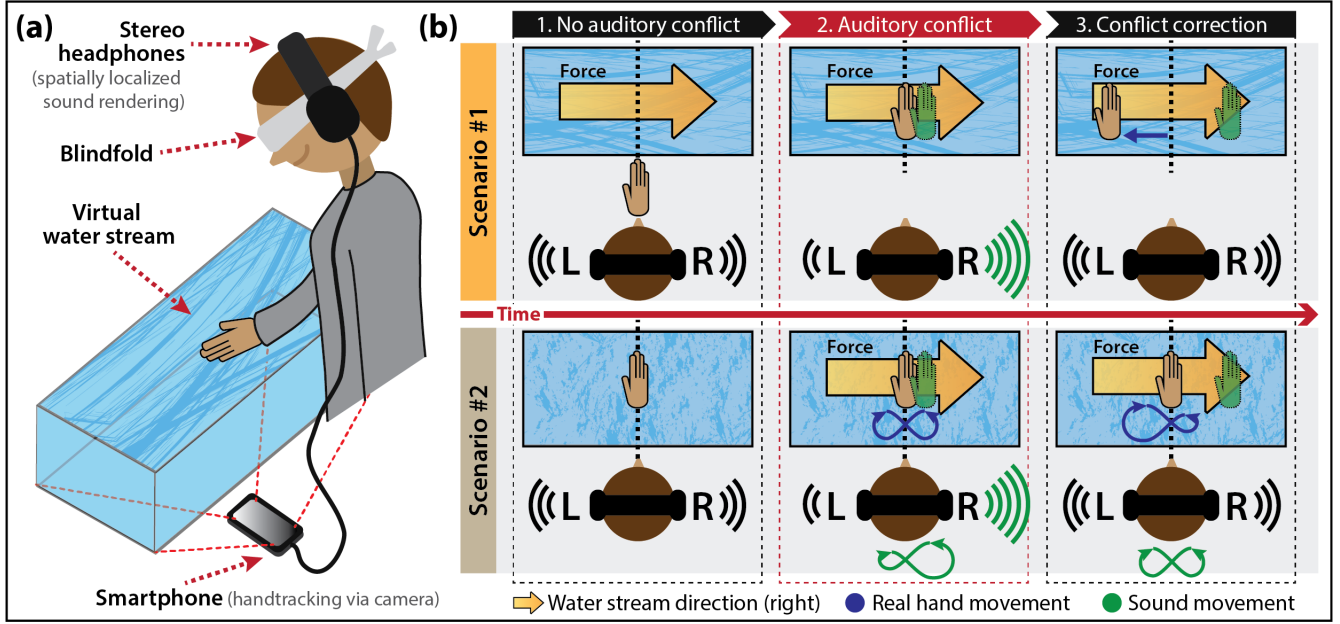


Figure 1: (a) System configuration (b) Water-stream scenarios to create pseudo-force feedback by shifting auditory hand position

Abstract

This proposal presents a pseudo-force feedback design based on spatially localized sound, tailored for the visually impaired. Sound location is adjusted to create an auditory conflict between the perceived hand position in virtual space and its actual position in the real world, thus inducing a force sensation. Two water-stream test scenarios were crafted in an Augmented Reality application built with MediaPipe for hand tracking and OpenAL for sound rendering.

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1 Introduction

The ongoing development of virtual immersive environments has led to a quest for solutions to provide accessible experiences, especially for the visually impaired. According to [Mott et al. 2019], with virtual reality technologies nearing maturity, it is vital to adopt accessibility as a core design principle for more inclusive experiences. They also discuss that visually impaired people cannot enjoy most virtual reality experiences because these mainly rely on visuals (ibid). In this context, literature presents non-visual alternatives to convey the extensive visual information in these environments [W3C RQTF 2017]. One such alternative is haptics, promoted to offer more realistic and consistent experiences for visually impaired people in this domain [Mott et al. 2019; W3C RQTF 2017].

In the realm of haptics, haptic and pseudo-haptic technologies both aim to induce tactile sensations. While haptic technologies provide direct stimuli using physical actuators, pseudo-haptics produce haptic illusions by stimulating other senses [Melaisi et al. 2018; Ujitoko and Ban 2021]. Haptic devices offer more realistic tactile experiences, but their complexity is often impractical, making pseudo-haptic a more feasible approach [Ujitoko and Ban 2021].

Various pseudo-haptic designs have been explored, but most are not suitable for the visually impaired. This is because pseudo-haptic

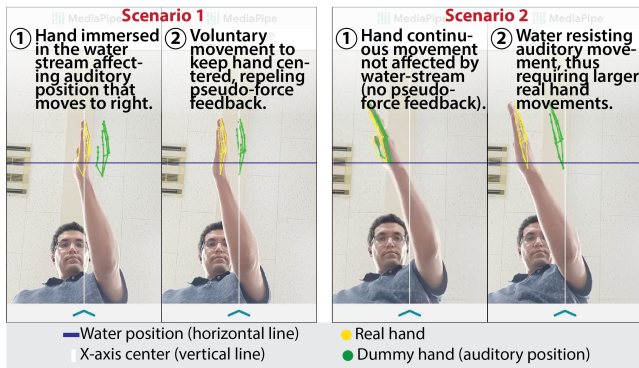


Figure 2: Water stream scenarios in mobile application

designs mainly rely on visual stimuli, leaving sound-based pseudo-haptics virtually unexplored [Melaissi et al. 2018; Ujitoko and Ban 2021]. However, recent studies aim to induce haptic sensations using sound stimuli [Bosman et al. 2021; Melaissi et al. 2018]. Similarly, the purpose of this study is to explore sound spatial localization and other parameters to create a pseudo-haptic force sensation.

2 Method

This research presents a pseudo-haptic design built on modalities for the visually impaired, which alters perceived hand position through sound modulation to create a force sensation. This solution is implemented as an Augmented Reality application in Java/C++ on an Android device, having mid-air hand gestures as main input, and spatially localized sounds delivered through stereo headphones as main output (see Fig. 1). Hand gestures are recognized via the device's camera using hand tracking feature of MediaPipe library, while sound stimuli are rendered using the OpenAL library, which uses Head-Related Transfer Function (HRTF) for spatial localization.

The force illusion arises from a conflict between the perceived "auditory position" of the hand in virtual space and its actual position in the real world. For instance, if the auditory position shifts involuntarily to the right, the user perceives a force pushing their hand, leading them to move it left to repel that sensation. This experience features two water-stream scenarios where auditory cues simulate water pushing the user's hand. In Scenario 1, users immerse their hand in a virtual river stream and try to keep it centered, while the auditory position is programmatically adjusted to simulate the force of the water. In Scenario 2, users place their hand in a virtual water tank and sweep it left and right to keep water flowing, while the auditory position is tuned to reflect the force exerted by underwater streams that assist or resist hand movement.

The system implements slightly different sound interactions across scenarios but operates similarly to produce the force illusion. Both scenarios position virtual water in front of the user; Scenario 1 conveys its presence with a river-like sound that intensifies with the hand's proximity (Z-axis), while Scenario 2 has no presence sound. In both cases, a splash sound occurs when the hand enters (or exits) the water, altering the gain and pitch of the river-like sound in Scenario 1 and initiating an underwater-like sound in Scenario 2. Inside the water, the system ignores front-to-back hand motions, focusing solely on lateral movements (X-axis), which change water sound

location to convey hand's auditory position. Scenario 1 features a wider central area unaffected by sound localization for seamless left-to-right transitions, while Scenario 2 lacks a defined center, resulting in more distinct transitions. Both scenarios produce the force illusion by shifting hand's auditory position laterally at a controlled velocity for a set period, with attenuation occurring as it moves further from the center, simulating the hand moving away and compelling the user to correct the involuntary displacement.

System features such as water position, center coordinates, and attenuation model, along with sonic parameters like gain, pitch, and velocity, were fine-tuned through preliminary tests for authenticity.

3 Insights from preliminary tests

During development, various parameters and system features were found to be critical for ensuring optimal auditory interaction in the virtual environment, thus requiring rigorous testing and refinement.

Environmental parameters affecting the entire virtual space included the listener's position ($[x, y, z] = [0, 0, 0]$) and orientation ($[x_{at}, y_{at}, z_{at}, x_{up}, y_{up}, z_{up}] = [0, 0, 1, 0, 1, 0]$), configured to enable first-person interactions. Moreover, the attenuation model (Inverse Clamped), reference distance (1.5 units), and maximum distance (35 units) parameters were fine-tuned to ensure realistic sound attenuation as sources moved farther from the listener's position.

Water-sound parameters, which varied with user interaction, were carefully adjusted to provide effective auditory feedback for the force illusion. Key parameters included sound velocity, gain, pitch, and frequency filters. Conversely, reverberation was detrimental to sound spatial localization and was hence omitted.

Additionally, certain system features were essential for enabling the required interaction in each scenario. In Scenario 1, defining a center area range ($-2 \leq x \leq 2$) and meticulously positioning the water stream along the Z-axis ($z = 3$) were necessary for smooth lateral transitions and perceivable slight lateral movements. Moreover, Scenario 1 needed a fixed velocity (0.6 units/s) to convey water stream force, while Scenario 2 required a dynamic velocity ($v = 0.07$) applied to the computed distance from hand movement rate. Similarly, a time slot for the force illusion was set to a minimum duration (10 s) for user comprehension. However, further testing revealed the need for a shift correction threshold (3 units) in Scenario 1 to end the force illusion early if the real hand's movement became too noticeable, ensuring the illusion remained believable.

References

- Isak de Villiers Bosman, Koos De Beer, and Theo J D Bothma. 2021. Creating pseudo-tactile feedback in virtual reality using shared crossmodal properties of audio and tactile feedback. *S. Afr. Comput. J.* 33, 1 (July 2021).
- Mohammed Melaissi, David Rojas, Bill Kapralos, Alvaro Uribe-Quevedo, and Karen Collins. 2018. Multimodal Interaction of Contextual and Non-Contextual Sound and Haptics in Virtual Simulations. *Informatics* 5, 4 (2018). <https://doi.org/10.3390/informatics5040043>
- Martez Mott, Edward Cutrell, Mar Gonzalez Franco, Christian Holz, Eyal Ofek, Richard Stokley, and Meredith Ringel Morris. 2019. Accessible by Design: An Opportunity for Virtual Reality. In *2019 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. 451–454. <https://doi.org/10.1109/ISMAR-Adjunct.2019.00122>
- Yusuke Ujitoko and Yuki Ban. 2021. Survey of Pseudo-Haptics: Haptic Feedback Design and Application Proposals. *IEEE Transactions on Haptics* 14, 4 (2021), 699–711. <https://doi.org/10.1109/TOH.2021.3077619>
- W3C RQTF. 2017. *Accessibility of Virtual Reality*. W3C Web Accessibility Initiative (WAI). Retrieved May 10, 2024 from https://www.w3.org/WAI/APA/task-forces/research-questions/wiki/Accessibility_of_Virtual_Reality